European Journal of Preventive Cardiology (2024) 31, 1595–1605
https://doi.org/10.1093/eurjpc/zwae156

FULL RESEARCH PAPER
Cerebrovascular disease & stroke

Age at menopause and risk of ischaemic stroke: a systematic review and meta-analysis

Jie Wang and Ying Chen *

Department of Obstetrics and Gynecology, Beijing Tiantan Hospital, Capital Medical University, 119 South Fourth Ring Road West, Fengtai District, Beijing 100070, China

Received 27 November 2023; revised 5 March 2024; accepted 20 April 2024; online publish-ahead-of-print 3 May 2024

Aims Despite ischaemic stroke having much importance as one of the top 10 causes of death in older women, there are limited data on age at menopause and ischaemic stroke. We performed a systematic review and meta-analysis to estimate the effect of age at menopause on ischaemic stroke.

Methods and results We screened four databases (PubMed, Cochrane, Web of Science, and EMBASE databases) up to 17 July 2023. This systematic review was reported in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-analyses (PRISMA) guidelines and was registered with PROSPERO (CRD42023444245). Data extraction and quality assessment were independently undertaken by two reviewers. A random-effects model was used for meta-analysis using Revman5.4 to calculate the risk ratio of the incidence of ischaemic stroke. Heterogeneity was assessed using I^2. Meta-regression and assessment for bias were performed. Out of 725 records identified, 10 studies were included in the qualitative synthesis and the quantitative meta-analysis. The pooled incidence rate for ischaemic strokes which age at menopause before 43 years old was 1.22 [95% confidence interval (CI): 1.02–1.46]. The pooled incidence rate of early menopause was 1.26 (95% CI: 1.07–1.48) for ischaemic stroke. The incidence rate of ischaemic stroke for women with early menopause may be in an environment with a high incidence for a long time.

Conclusion This meta-analysis suggests that early menopause is associated with an increased risk of ischaemic stroke. Age at onset of menopause before 43 years old may be the cut-off value of increased risk of ischaemic stroke.

Lay summary Ischaemic stroke has much importance as one of the top 10 causes of death in older women. We performed a systematic review and meta-analysis to estimate the effect of age at onset of menopause on ischaemic stroke. Our findings suggest that early menopause is associated with an increased risk of ischaemic stroke. Age at onset of menopause before 43 years old may be the cut-off value of increased risk of ischaemic stroke, which provides some nexus between the relationship of early menopause and ischaemic stroke risk.

Keywords Menopause • Cerebrovascular disorders • Ischaemic stroke • Incidence • Systematic review • Meta-analysis

Introduction

The risk of cardiovascular and cerebrovascular diseases (CCVDs) increases with ageing, and because women tend to live longer than men, more women suffer from and die from CCVD.[1] Therefore, early identification of women at high risk of CCVD and timely implementation of appropriate lifestyle or therapeutic interventions are of public health importance. As the most common manifestation of cerebrovascular disease, stroke is the fourth leading cause of death and a major cause of severe disability.[2,3] Ischaemic stroke could result from multiple events such as embolism with a cardiac origin, occlusion of small vessels in the brain, and atherosclerosis affecting the cerebral circulation. Ischaemic stroke is the most important causes of neurological morbidity and mortality.[4]

The risk for ischaemic stroke was significantly lower in midlife women than men, but this difference diminished with age.[5,6] Currently, known risk factors for stroke, such as hypertension, smoking, and ischaemic heart disease, are more common in men.[7] However, they can only partly explain the differences in stroke incidence. Furthermore, little attention has been paid to the influence of sex-specific risk factors on the risk of ischaemic stroke.

* Corresponding author. Tel: 0086-010-59976651, Email: cy.doc@163.com

Table 1 The Newcastle–Ottawa scale for observational studies

The Newcastle–Ottawa Scale for cohort studies

Study	Selection				Comparability	Outcome			Quality score	Overall quality
	Representativeness of the exposed cohort	Selection of the non exposed cohort	Ascertainment of exposure	Demonstration that outcome of interest was not present at the start of study	Comparability of cohorts on the basis of the design or analysis	Assessment of outcome	Was follow-up long enough for outcomes to occur	Adequacy of follow-up of cohorts		
Su 2023	★	★	★	★	★	★	★	★	8	High
Lisabeth 2009	★	★	★		★★	★	★	★	8	High
Hu 2021	★	★	★		★★	★	★	★	8	High
Lena 2023	★	★	★	★	★★	★	★	★	9	High
Choi 2005		★	★	★	★	★	★	★	7	High
Michael 2019	★	★	★		★★	★	★	★	8	High
Baba 2010	★	★	★		★	★	★	★	7	High
Hu 1999		★	★	★	★	★	★	★	7	High

The Newcastle–Ottawa Scale for case–control studies

Study	Selection				Comparability	Exposure			Quality score	Overall quality
	Is the case definition adequate	Representativeness of the cases	Selection of controls	Definition of controls	Comparability of cases and controls on the basis of the design or analysis	Ascertainment of exposure	Same method of ascertainment for cases and controls	Non-response rate		
Alonso 2007	★	★	★	★	★	★	★		7	High
Hsieh 2010	★	★		★	★★	★	★	★	8	High

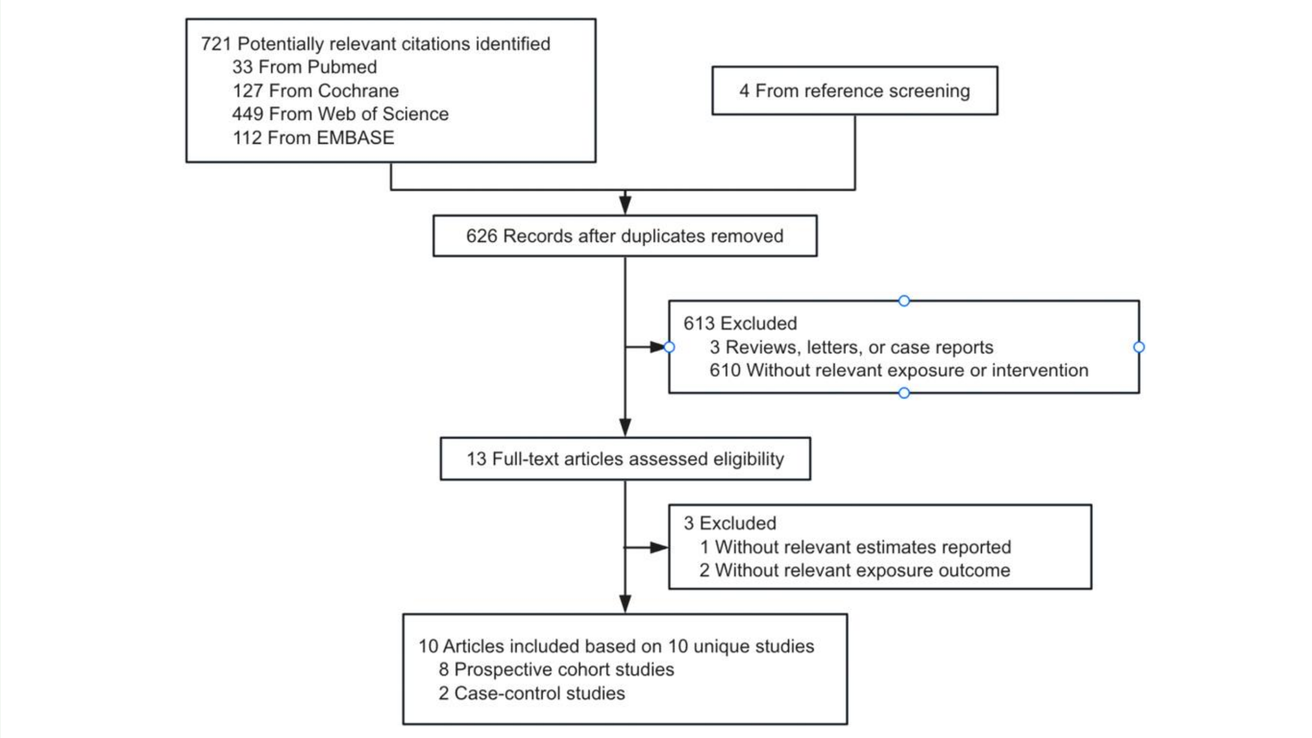

Figure 1 PRISMA flow diagram of the selection process and inclusion of studies.

Female oestrogen participates in the relaxation and dilation of blood vessels, accommodating blood flow and activating the renal–angiotensin–aldosterone system. Therefore, reduced oestrogen levels after menopause not only harden the blood vessels,[8] but also lead to endothelial dysfunction, inflammation, and immune dysfunction.[9] After menopause, the endogenous oestrogen level decreases, and the risk of CCVD increases dramatically, making menopause as a high-risk period for CCVD in women.[10,11] Within the first 10 years after menopause, a woman's risk for ischaemic stroke roughly doubles.[12] Therefore, the importance of cardiovascular and cerebrovascular risk assessment and prevention should be emphasized during this period.

The age at onset of menopause may be a marker not only of reproductive ageing but also of general health and physical ageing.[13] Early menopause may affect the risk of ischaemic stroke, and women with early menopause may be in an environment with a high incidence of CCVD for a long time.[14] However, the exact relationship between age at menopause and ischaemic stroke is still unclear. This article intends to focus on the effect of age at onset of menopause on ischaemic stroke.

We performed a systematic review and meta-analysis of the available observational evidence to quantify the association of age at menopause with the occurrence of ischaemic stroke.

Methods

Search strategy and selection criteria

We conducted a systematic review of published articles from the first relevant article published in 1989 up to 17 July 2023, using the PubMed, Cochrane, Web of Science, and EMBASE databases. The search was restricted to studies published in English and human studies using the following search keywords and medical subject heading terms in the PubMed (MeSH), Cochrane, Web of Science, and EMBASE (Emtree): (menopause OR menopausal OR post-menopause OR postmenopause OR post-menopausal OR postmenopausal) AND ("Ischemic stroke 'OR' Ischemic strokes 'OR' Ischemic cerebrovascular disorder"). We further reviewed the references of all included articles to identify any additional studies not identified by our search strategy (known as the snowballing approach). This systematic review was reported in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-analyses (PRISMA) guidelines and was registered with PROSPERO (CRD42023444245).

J.W. and Y.C. screened the titles and abstracts of identified articles and reviewed the full text of relevant articles. Articles that meet our inclusion criteria are observational studies (i.e. cohort or case–control study design) reporting age at natural menopause (surgical and hypothalamic–pituitary amenorrhoea are excluded) and examining the association with ischaemic stroke. Studies were excluded if they were review papers, commentaries, conference abstracts, and studies without relevant exposures, outcomes, or effect estimates [i.e. relative risk (RR), hazard ratio, or odds ratio]. We contacted original authors by e-mail to obtain missing data. Excel2021 was used to tabulate and Revman5.4 was used to visually display results of individual studies and syntheses. The review protocol can be accessed from Supplementary material online.

Data extraction

Data presented in tables and figures were derived directly from published findings. All data extracted in this paper are directly available from the original literature. Data extraction and quality assessments were conducted by two independent reviewers (J.W. and Y.C.), and any disagreement was resolved by discussion. The following information was extracted from the eligible articles: first author, date of publication (year), study title, year of baseline survey, location of study, mean age of baseline, length of follow-up (year), sample size and cases, study participants, menopausal status, categorical age at menopause, and ascertainment of ischaemic stroke. The multivariable-adjusted RRs with 95% confidence intervals (CIs) were extracted for the meta-analysis.

Table 2 Characteristics of 10 included studies

Lead author, publication date	Address of the first author	Location	Year of baseline survey	Baseline age range/average age (years)	Study design	Follow-up years (SD)	Total participants	Outcome	Study participants	Study quality	Age at menopause (age categories) (years)	Definition of early menopause (years)
Su 2023	Department of Medicine, Seoul National University College of Medicine, Seoul, Republic of Korea	Korea	2009	66.9 ± 8.3/ 60.8 ± 8.0	Retrospective cohort study	8.4	61 067/ 1 163 480	Ischaemic stroke (IS)	Population-based retrospective cohort study from the National Health Insurance Service database of Korea	High	<40, 40–45, 46–50, 51–54, and ≥ 55	<45
Alonso 2007	Department of Neurology, University Hospital Ramon y Cajal, Ctra de Colmenar Km 9,100, 28034 Madrid, Spain	Spain	2001	68.9 (46–93)	Case–control study	3	430/905	Ischaemic stroke (IS)	Multicentre, age-matched case–control study conducted in 18 hospitals	High	<47, 47–50, 51–53, and ≥53	<47
Lisabeth 2009	Department of Epidemiology (L.D.L.), University of Michigan School of Public Health, Ann Arbor, MI, USA	USA	1984	67.2	Prospective cohort study	22 (9)	234/1430	Ischaemic stroke (IS)	A prospective cohort study of 5209 participants (2873 women, ages 28–62 at the time of enrolment) that began in 1948 in the town of Framingham, Mass	High	<42, 42–54, ≥ 55	<42
Hu 2021	Department of Internal Medicine and Central Laboratory, Guangzhou, China	China	2003	61.4	Retrospective cohort study	12	222/16 282	Ischaemic stroke (IS)	Prospective cohort study including permanent Guangzhou residents aged 50 years or older that aims to examine environmental and genetic determinants of chronic diseases	High	<43, 43–47, 48–50, 51–52, and ≥ 53	<43

Continued

Table 2 Continued

Lead author, publication date	Address of the first author	Location	Year of baseline survey	Baseline age range/ average age (years)	Study design	Follow-up years (SD)	Total participants	Outcome	Study participants	Study quality	Age at menopause (age categories) (years)	Definition of early menopause (years)
Lena 2023	Julius Center for Health Sciences and Primary Care, University Medical Center Utrecht, Utrecht, The Netherlands	The Netherlands	2006–2010/ 1992–2000	58.9 (SD = 5.8)	Cohort study	12.6	6770/189 095	Ischaemic stroke (IS)	UK Biobank and EPIC-CVD study	High	<40, 40–45, 45–50, 50–55, and ≥55	<45
Choi 2005	Department of Neurology, College of Medicine, Inha University, Incheon	Korea	1993	69.8 ± 5.5	Retrospective cohort study	5	93/5638	Ischaemic stroke (IS)	Korean Elderly Pharmacoepidemiologic Cohort	High	<40, 40–44, 45–49, and ≥55	<45
Michael 2019	Cardiology Division, Massachusetts General Hospital, Harvard Medical School, Boston, MA, USA	USA	2006–2010	40–69	Cohort study	10	9075/144 260	Ischaemic stroke (IS)	Adult residents of the UK recruited between 2006 and 2010. Of women who were 40–69 years old and postmenopausal at study enrolment	High	<40, ≥ 40	<40
Baba 2010	Department of Obstetrics and Gynecology, Haga Red Cross Hospital, Tochigi, Japan	Japan	1992–1995	61.0 (SD = 4.7)	Cohort study	10.8 (2.2)	80/3788	Ischaemic stroke (IS)	The Jichi Medical School (JMS) Cohort Study	High	<40, 40–44, 45–49, and ≥55	<45

Continued

Table 2 *Continued*

Lead author, publication date	Address of the first author	Location	Year of baseline survey	Baseline age range/average age (years)	Study design	Follow-up years (SD)	Total participants	Outcome	Study participants	Study quality	Age at menopause (age categories) (years)	Definition of early menopause (years)
Hu 1999	Department of Nutrition, Harvard School of Public Health, Boston, MA 02115, USA	USA	1976	30–55	Cohort study	18	350/354 326	Ischaemic stroke (IS)	The Nurses' Health Study cohort	High	<40, 40–44, 45–49, 50–54, and ≥55	<45
Hsieh 2010	School of Public Health, Taipei Medical University, Taipei	China	N/A	66.4 ± 9.5/64.9 ± 8.9	Case–control study	N/A	189/192	Ischaemic stroke (IS)	People were recruited from Chi Mei Medical Center, Lotung Poh-Ai Hospital, Wan-Fang Hospital, and Taipei Medical University Hospital	High	≤48,49–50, ≥51	<49

Quality assessment

The Newcastle–Ottawa Scale for observational studies was used to assess the risk of bias for the included articles. Selection, comparability, and outcome/exposure assessment were rated separately for cohort and case–control studies, and the rating scores ranged from 0 to 9 (highest to lowest degree of bias). Scores of 0–3, 4–6, and 7–9 were classified as low, moderate, and high quality, respectively (*Table 1*). Begg's funnel plots were adopted to examine the publication bias for all primary and secondary outcomes defined above.

Statistical analysis

Random-effects meta-analysis was conducted to synthesize results from the included studies. We specifically examined the risk of ischaemic stroke associated with age at menopause and separately examined the risk of ischaemic stroke using early and late menopause as reference categories. All analyses were conducted using log-incidence rates and log-standard errors derived from the 95% CIs. In the meta-analysis, the distribution of incidence rates and 95% CIs obtained were examined using a forest plot. Pooled estimates of the incidence rate were used to determine an overall rate using a random-effects model. Assessment of heterogeneity was performed using the I^2 statistic and the test of heterogeneity (Q-statistic); values of $I^2 > 50\%$ and $P < 0.05$ were considered significant heterogeneity between studies.[15]

Subgroup analyses were conducted to explore potential causes of heterogeneity by publication year, study setting, study design, and overall quality assessment score. A test of group differences (Cochran's Q) was used to test for the presence of heterogeneity between groups.

Results

Identification of relevant articles

This search strategy identified 721 articles from PubMed, Cochrane, Web of Science, and EMBASE databases (*Figure 1*). Four additional eligible articles were identified using the snowball method (reference screening). After removing 99 duplicate articles, 626 articles were screened based on title and abstract. Of these, 13 articles met the criteria for full-text review. Three articles are excluded, one article did not have estimates reported and another two articles did not have relevant outcome. A total of 10 articles were selected according to the outcome events. A comprehensive review of the literature was conducted according to the inclusion and exclusion criteria, and a total of 10 articles were included (based on 10 independent studies).

Study characteristics

Of the 10 included studies, 8 were cohort studies, 2 were case–control studies (*Table 2*). Study populations were drawn from six countries (three in Asia, two in Europe, and one in the America), with the mean age of baseline ranging from 30 to 75 years (most were postmenopausal) and mean follow-up years ranging from 3 to 22 years.

Age at menopause and risk of ischaemic stroke

Cohort studies and case–control studies were analysed as two subgroups in this meta-analysis because of the bias in the level of evidence, sampling error, and randomness between them.

Eight large cohort studies showed an increased risk of ischaemic stroke with early menopause (RR 1.32, 95% CI 1.08–1.60; I^2: 95%, $P < 0.00001$). Conversely, two large cohort studies showed no increased risk of ischaemic stroke with early menopause (RR 1.14, 95% CI 0.99–1.31; I^2: 0%, $P = 0.73$). Compared with case–control studies, cohort studies have a higher level of evidence due to reliable data sources, so the credibility of the results of cohort studies is higher. When the two subgroups of cohort and case–control studies were combined, the results showed that early menopause increased the risk of ischaemic stroke (RR 1.26, 95% CI 1.07–1.48; I^2: 90%, $P < 0.00001$) (*Figure 2A*).

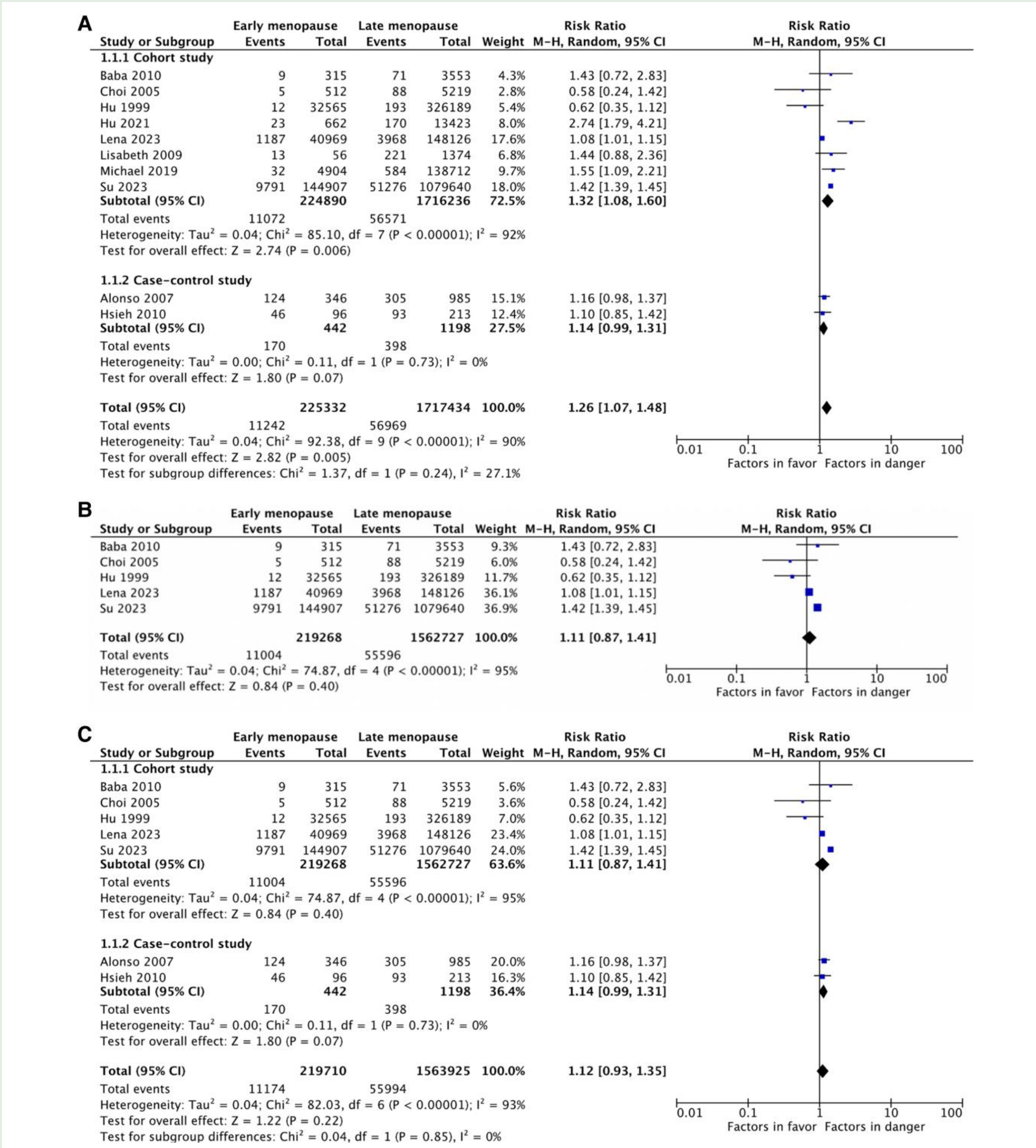

Figure 2 Risk of ischaemic stroke for early menopausal women. (*A*) The summary estimates for meta-analysis of the association between age at onset of menopause and ischaemic stroke. (*B*) Age at onset of menopause before 45 years old and the risk of ischaemic stroke. (*C*) Age at onset of menopause before 49 years old and the risk of ischaemic stroke. (*D*) Age at onset of menopause before 43 years old and the risk of ischaemic stroke. (*A*) There was evidence of between-study heterogeneity for age at menopause. Across all included studies, early menopause was associated with an increased risk of ischaemic stroke (relative risk 1.26, 95% confidence interval 1.07–1.48; I^2: 90%, $P < 0.00001$). There was a significant heterogeneity between groups ($P = 0.005$), and the group with women age at onset of menopause before 43 years contributed 24.5% to the weight. (*B*) There was evidence of between-study heterogeneity for age at menopause younger than 45 (relative risk 1.11, 95% confidence interval 0.87–1.41; I^2: 95%, $P < 0.00001$). (*C*) There was evidence of between-study heterogeneity for age at menopause younger than 49 (relative risk 1.12, 95% confidence interval 0.93–1.35; I^2: 95%, $P < 0.00001$). (*D*) There was evidence of between-study heterogeneity for age at menopause younger than 43 (relative risk 1.22, 95% confidence interval 1.02–1.46; I^2: 92%, $P < 0.00001$).

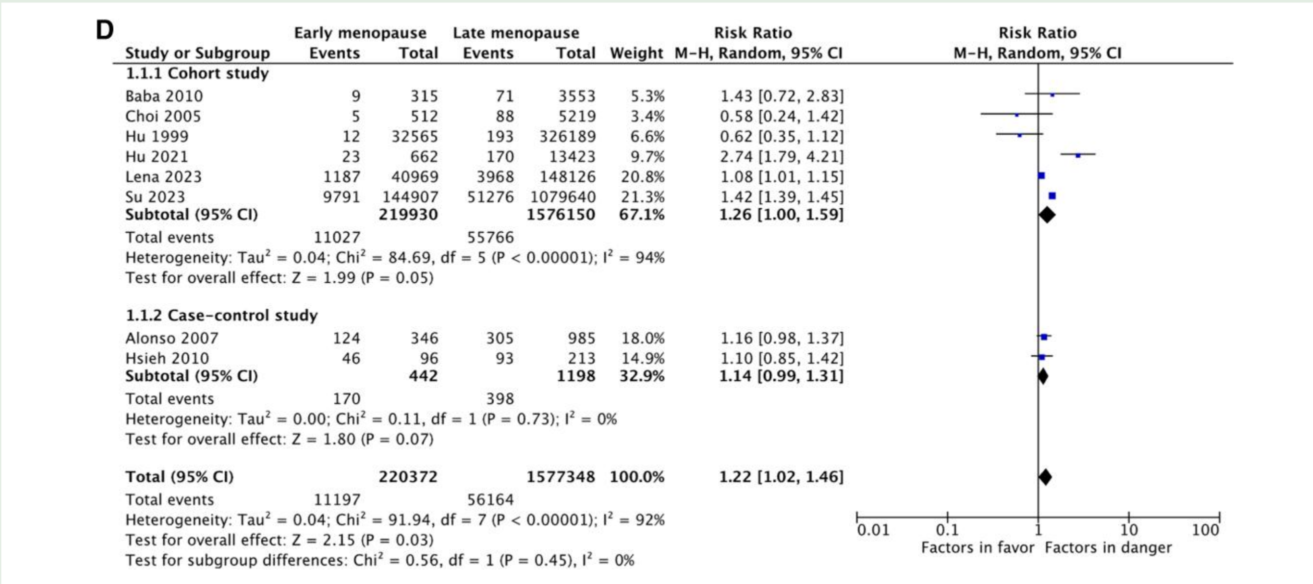

Figure 2 Continued

Figure 2 shows the summary estimates from the meta-analysis of the association between age at menopause and ischaemic stroke. Of these 10 studies, 5 studies defined menopausal age before 45 years old as early menopause.[16–20] The other five studies defined menopausal age before 49, 47, 43, 42, and 40 years old as early menopause, respectively[21–25] (*Table 2*). For the age at menopause and the incidence of ischaemic stroke, *Figure 2B–D* demonstrates the risk of menopause and ischaemic stroke for women whose age at menopause was before 45, 49, and 43 years, respectively.

There are two articles, one defines menopause age before 43 years old as early menopause[23] and the other defines menopause age before 49 years old as early menopause.[26] Considering that there are cases of premature menopause due to iatrogenic factors or delayed menopause due to the use of hormone replacement therapy in the age at menopause group between 43 and 49 years old, there may be both early menopause and delayed menopause relative to the body's physiological menopause age. Therefore, we considered both the population of age at menopause before 43 and 49 years old as early menopause groups and analysed the heterogeneity of the articles (*Figure 2C* and *D*).

Age at onset of menopause before 45 or 49 years old was not associated with an increased risk of ischaemic stroke (RR 1.11, 95% CI 0.87–1.41; I^2: 95%, $P < 0.00001$; RR 1.12, 95% CI 0.93–1.35; I^2: 93%, $P < 0.00001$, respectively). In the cohort study subgroup, age at menopause before 43 years old was associated with a higher risk of ischaemic stroke (RR 1.26, 95% CI 1.00–1.59; I^2: 94%, $P < 0.00001$), although the risk of ischaemic stroke was not significantly higher in the subgroup of case–control studies (RR 1.14, 95% CI 0.99–1.31; I^2: 0%, $P = 0.73$), but with a trend towards a higher risk of ischaemic stroke. Age at onset of menopause before 43 years old was associated with an increased risk of ischaemic stroke (RR 1.22, 95% CI 1.02–1.46; I^2: 92%, $P < 0.00001$).

Sensitivity analysis and publication bias

Funnel plots for the risk of ischaemic stroke by menopausal age are shown in *Figure 3A–C*. As shown in the funnel plot, the results of these studies are basically symmetrically distributed on both sides of the null line, the number of studies with positive results is basically equal to the number of studies with negative results, and the figure shows a clear symmetrical distribution, indicating that publication bias does not exist, or if there is bias, it is small.

Discussion

Relevant scholars believe that ischaemic stroke is quite rare in premenopausal women, but the risk increases with age.[27] Menopause refers to the permanent cessation of menstruation, which represents a permanent oestrogen deficiency. Studies of the association between age at onset of menopause and risk of ischaemic stroke are rare and somewhat inconsistent in findings. Hu *et al.*[20] found in the Nurses' Health study that the age at natural menopause was not associated with the risk of ischaemic stroke in individuals who had never used hormone therapy. In a cohort study of postmenopausal Korean women who had never used hormone therapy, Choi *et al.*[18] found no association between age at natural menopause and the risk of ischaemic stroke, On the contrary, observational studies[28] have shown that compared with men of the same age, young, premenopausal women are less susceptible to ischaemic stroke, and this protection may fade away with age. Using data from the Framingham Heart Study, Lisabeth *et al.*[24] found that women who had natural menopause before 42 years old had twice the risk of ischaemic stroke compared with women who had natural menopause at the age of 42 and older. Results from a Japanese cohort study[19] also suggest that after adjusting for age and risk factors, women who experienced menopause before 40 years old were more than twice as likely to have ischaemic stroke as women who experienced menopause between age of 50 and 54; however, this study included both natural and surgical menopausal women.

For Chinese women, the average age to enter perimenopause is 46 years old, the average age at onset of menopause is 48–52 years old, and ∼90% of women go through the menopause between the ages of 45 and 55. Early menopause is defined as menopause between the ages of 40 and 45.[29] Although the vast majority of women will follow the law of natural ovarian ageing, there are still a few women who experience ovarian function decline or even failure before the age of 40. About 1% of women underwent premature menopause before the age of 41, and 3% of women underwent menopause between the ages of 41 and 46.[30] Early menopause can affect all systems of women, but

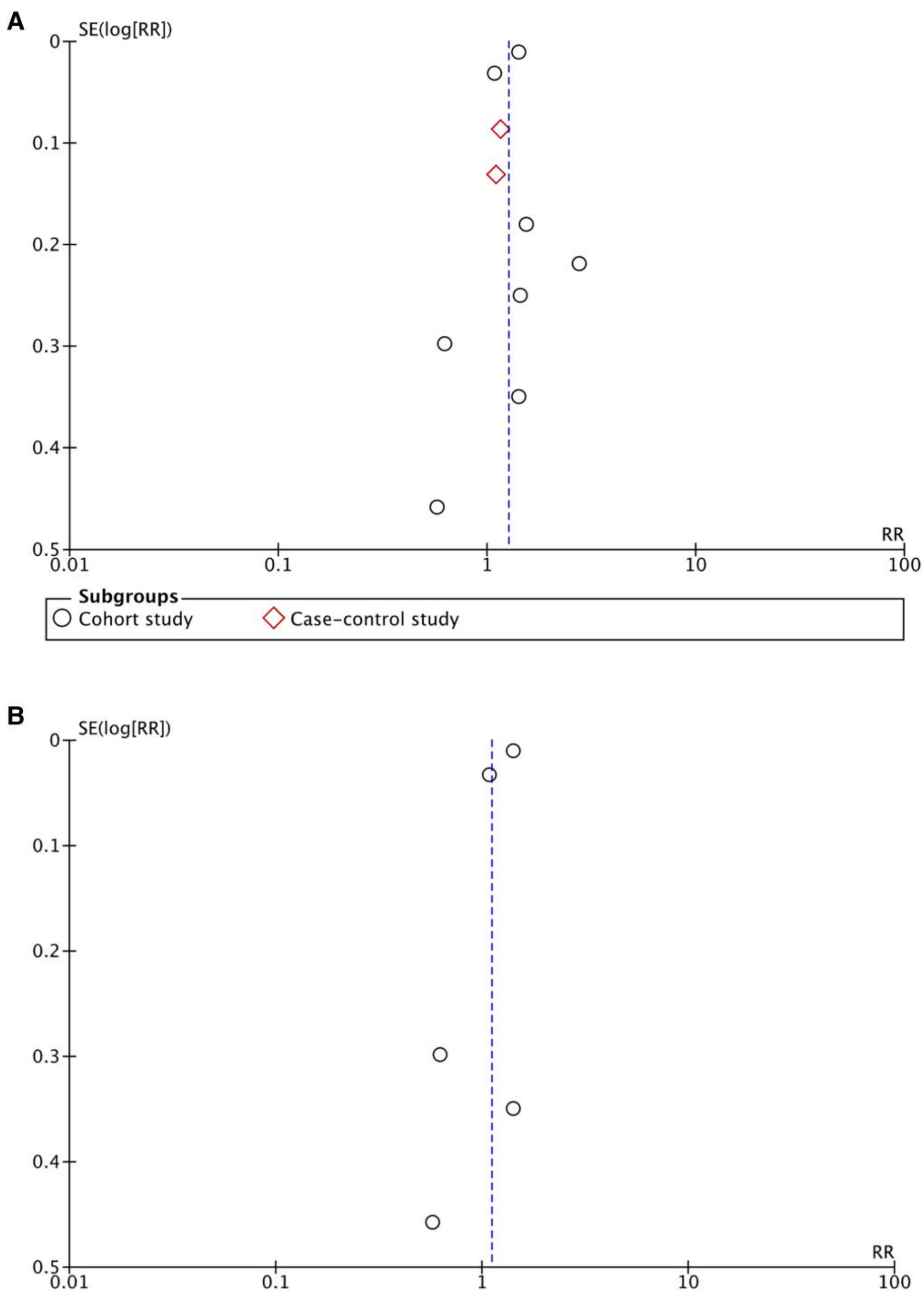

Figure 3 Funnel plots of the risk of ischaemic stroke for early menopausal women. (*A*) Funnel plots of summary estimates of meta-analyses between early menopause and risk of ischaemic stroke. (*B*) Funnel plots of the risk of the age at onset of menopause before 45 years old and ischaemic stroke. (*C*) Funnel plots of the risk of age at onset of menopause before 49 years old and ischaemic stroke. (*D*) Funnel plots of the risk of age at onset of menopause before 43 years old and ischaemic stroke.

for the occurrence of diseases in different systems, the effects of different age at menopause are also different. Muka *et al*.[31] showed that women whose age at menopause before 45 have a higher risk of coronary heart disease, cardiovascular disease mortality, and total mortality.

For the occurrence of ischaemic stroke, different from the established concept of early menopause in people whose age at menopause before 45 years old, our study showed that people whose age at menopause before 43 years old have higher risk. There was no difference in the incidence of ischaemic stroke between those whose age at onset of menopause before 45 years old and those whose onset of menopause before 49 years old. This suggests that the age at onset of menopause before 43 years old is a more accurate cut-off value that reflects the

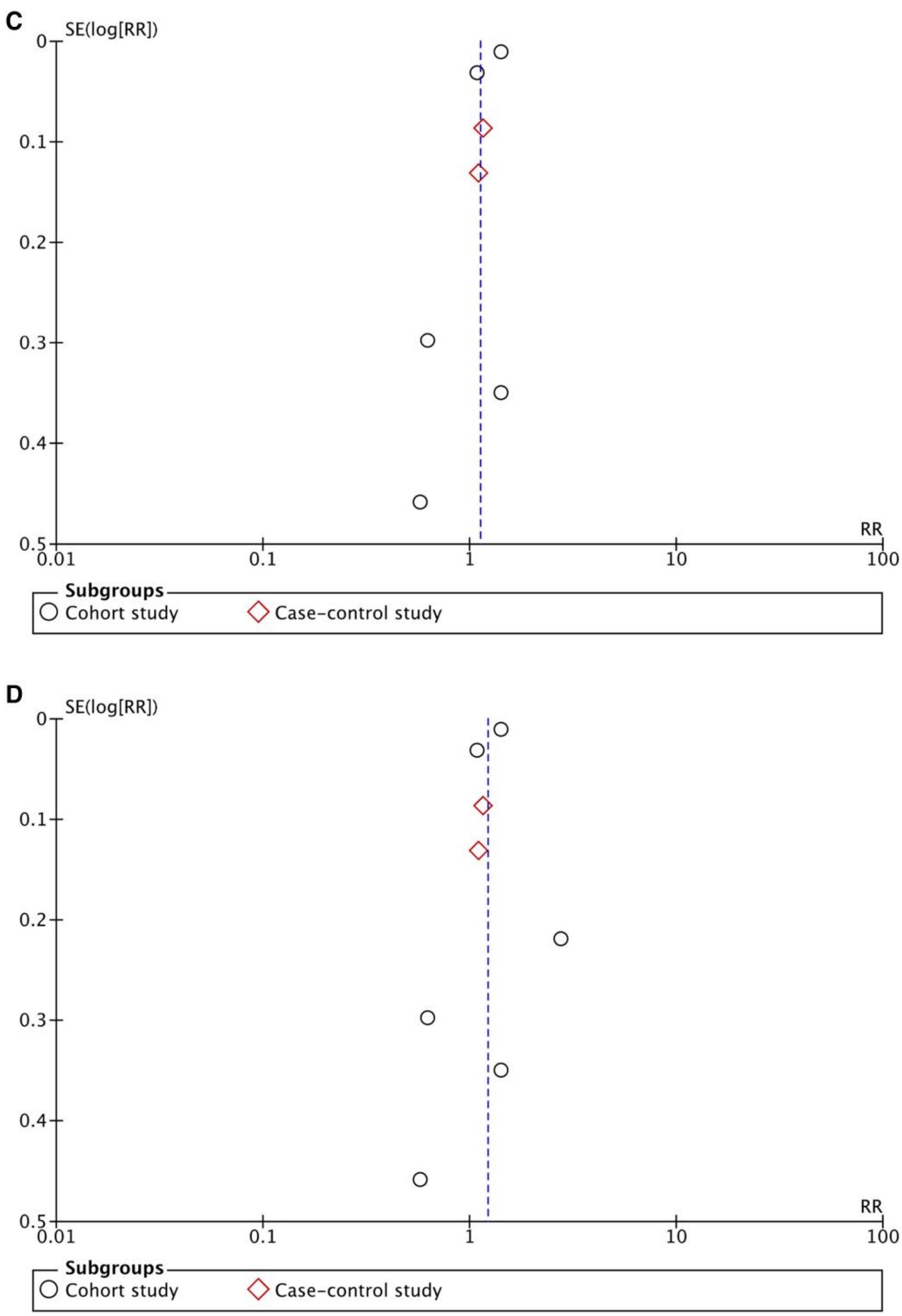

Figure 3 Continued

relationship between menopause and ischaemic stroke. There may be somewhat pathophysiological changes between 43 and 45 years of age in early menopausal women that make the different risk of ischaemic stroke, which is worthwhile to further investigate the mechanism.

While this study systematically reviewed 10 primary studies concerning the incidence of ischaemic stroke in early menopausal women, using recommended practice guidelines for selecting studies, it is not without limitations. The literature search was limited to studies reported in English. While this is a common practice in systematic reviews, we cannot rule out the possibility that relevant studies may have been missed due to this restriction. Owing to differential categorization of age at onset of menopause in various studies and the few studies evaluating a particular outcome (e.g. ischaemic stroke), some of our analysis are based on a small number of studies. Therefore, these results need to be interpreted with caution, particularly those that yielded moderate between-study heterogeneity estimates. Furthermore, the lack of studies evaluating risk in relation to time since the onset of menopause limited us from performing any meaningful quantitative synthesis using this exposure.

While acknowledging these limitations, this study has some notable strengths. It is the first study to comprehensively examine the incidence of ischaemic stroke in early menopausal women.

Conclusions

Evidence from this meta-analysis suggests that early menopause is associated with an increased risk of ischaemic stroke. Age at onset of menopause before 43 years old may be the cut-off value of increased risk of ischaemic stroke, which provides some nexus between the relationship of early menopause and ischaemic stroke risk.

Supplementary material

Supplementary material is available at *European Journal of Preventive Cardiology*.

Acknowledgements

We appreciate the patients of the original studies included in this meta-analysis for their active cooperation, and we appreciate the researchers of the original studies included in this meta-analysis for their accurate data collection and statistical analysis, which provide a reliable data source for this meta-analysis.

Author contribution

J.W. conceptualized the study, performed statistical analyses, and drafted the manuscript. J.W. and Y.C. contributed to the literature search, screening, and data extraction. Y.C. conceptualized the study and provided critical revision of the manuscript for important intellectual content. All authors read and approved the final manuscript.

Funding

The authors received the National Natural Science Foundation of China (Grant No. 30901598) financial support for the research, authorship, and/or publication of this article.

Conflict of interest: The authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Data availability

All data are incorporated into the article and its online Supplementary material.

References

1. Thompson RH, Snyder AE, Burt DR, Greiner DS, Luna MA. Risk screening for cardiovascular disease and diabetes in Latino migrant farmworkers: a role for the community health worker. *J Community Health* 2015;**40**:131–137.
2. Feske SK. Ischemic stroke. *Am J Med* 2021;**134**:1457–1464.
3. Roger VL, Go AS, Lloyd-Jones DM, Adams RJ, Berry JD, Brown TM, *et al.* Heart disease and stroke statistics—2011 update: a report from the American Heart Association. *Circulation* 2011;**123**:e18–e209.
4. Maida CD, Norrito RL, Daidone M, Tuttolomondo A, Pinto A. Neuroinflammatory mechanisms in ischemic stroke: focus on cardioembolic stroke, background, and therapeutic approaches. *Int J Mol Sci* 2020;**21**:6454.
5. Löfmark U, Hammarström A. Evidence for age-dependent education-related differences in men and women with first-ever stroke. Results from a community-based incidence study in northern Sweden. *Neuroepidemiology* 2007;**28**:135–141.
6. Algra AM, Klijn CJ, Helmerhorst FM, Algra A, Rinkel GJ. Female risk factors for subarachnoid hemorrhage: a systematic review. *Neurology* 2012;**79**:1230–1236.
7. Appelros P, Stegmayr B, Terént A. Sex differences in stroke epidemiology: a systematic review. *Stroke* 2009;**40**:1082–1090.
8. Mendelsohn ME, Karas RH. The protective effects of estrogen on the cardiovascular system. *N Engl J Med* 1999;**340**:1801–1811.
9. Zhao Z, Wang H, Jessup JA, Lindsey SH, Chappell MC, Groban L. Role of estrogen in diastolic dysfunction. *Am J Physiol Heart Circ Physiol* 2014;**306**:H628–H640.
10. Sutton-Tyrrell K, Wildman RP, Matthews KA, Chae C, Lasley BL, Brockwell S, *et al.* Sex-hormone-binding globulin and the free androgen index are related to cardiovascular risk factors in multiethnic premenopausal and perimenopausal women enrolled in the Study of Women Across the Nation (SWAN). *Circulation* 2005;**111**:1242–1249.
11. Sutton-Tyrrell K, Lassila HC, Meilahn E, Bunker C, Matthews KA, Kuller LH. Carotid atherosclerosis in premenopausal and postmenopausal women and its association with risk factors measured after menopause. *Stroke* 1998;**29**:1116–1121.
12. Lisabeth L, Bushnell C. Stroke risk in women: the role of menopause and hormone therapy. *Lancet Neurol* 2012;**11**:82–91.
13. Snowdon DA, Kane RL, Beeson WL, Burke GL, Sprafka JM, Potter J, *et al.* Is early natural menopause a biologic marker of health and aging? *Am J Public Health* 1989;**79**:709–714.
14. Rocca WA, Grossardt BR, Miller VM, Shuster LT, Brown RD Jr. Premature menopause or early menopause and risk of ischemic stroke. *Menopause* 2012;**19**:272–277.
15. Nasser M. Cochrane handbook for systematic reviews of interventions. *Am J Public Health* 2020;**110**:753–754.
16. Jeong SM, Yoo JE, Jeon KH, Han K, Lee H, Lee DY, *et al.* Associations of reproductive factors with incidence of myocardial infarction and ischemic stroke in postmenopausal women: a cohort study. *BMC Med* 2023;**21**:64.
17. Tschiderer L, Peters SAE, van der Schouw YT, van Westing AC, Tong TYN, Willeit P, *et al.* Age at menopause and the risk of stroke: observational and Mendelian randomization analysis in 204 244 postmenopausal women. *J Am Heart Assoc* 2023;**12**:e030280.
18. Choi SH, Lee SM, Kim Y, Choi N-K, Cho YJ, Park B-J, *et al.* Natural menopause and risk of stroke in elderly women. *J Korean Med Sci* 2005;**20**:1053–1058.
19. Baba Y, Ishikawa S, Amagi Y, Kayaba K, Gotoh T, Kajii E. Premature menopause is associated with increased risk of cerebral infarction in Japanese women. *Menopause* 2010;**17**:506–510.
20. Hu FB, Grodstein F, Hennekens CH, Colditz GA, Johnson M, Manson JE, *et al.* Age at natural menopause and risk of cardiovascular disease. *Arch Intern Med* 1999;**159**:1061–1066.
21. Hsieh Y-C, Hwang L-C, Hsieh F-I, Lien L-M, Lin H-J, Hu C-J, *et al.* Early menarche and ischemic stroke risk among postmenopausal women. *Int J Gerontol* 2010;**4**:16–22.
22. Alonso de Leciñana M, Egido JA, Fernández C, Martínez-Vila E, Santos S, Morales A, *et al.* Risk of ischemic stroke and lifetime estrogen exposure. *Neurology* 2007;**68**:33–38.
23. Hu ZB, Lu ZX, Zhu F. Age at menarche, age at menopause, reproductive years and risk of fatal stroke occurrence among Chinese women: the Guangzhou Biobank Cohort Study. *BMC Womens Health* 2021;**21**:433.
24. Lisabeth LD, Beiser AS, Brown DL, Murabito JM, Kelly-Hayes M, Wolf PA. Age at natural menopause and risk of ischemic stroke: the Framingham Heart Study. *Stroke* 2009;**40**:1044–1049.
25. Honigberg MC, Zekavat SM, Aragam K, Finneran P, Klarin D, Bhatt DL, *et al.* Association of premature natural and surgical menopause with incident cardiovascular disease. *JAMA* 2019;**322**:2411–2421.
26. Sun J, Shi C, Liang Y, Niu J, Guo S, Cheng Z. Effects of early second-look hysteroscopy combined with intrauterine balloon dilatation on reproductive outcomes for women with intrauterine adhesions. *Int J Gynaecol Obstet* 2020;**149**:192–196.
27. Petrea RE, Beiser AS, Seshadri S, Kelly-Hayes M, Kase CS, Wolf PA. Gender differences in stroke incidence and poststroke disability in the Framingham Heart Study. *Stroke* 2009;**40**:1032–1037.
28. Lee JS, Yaffe K, Lui LY, Cauley J, Taylor B, Browner W, *et al.* Prospective study of endogenous circulating estradiol and risk of stroke in older women. *Arch Neurol* 2010;**67**:195–201.
29. Menopause Group, Society of Obstetrics and Gynecology, Chinese Medical Association. Chinese guideline for menopausal management and menopausal hormone therapy (2023 edition). *Chin J Obstet Gynecol* 2023;**58**:4–21.
30. Bullington BW, Edmonds A, Ramirez C, Rahangdale L, Neal-Perry G, Konkle-Parker D, *et al.* Premature and early menopause among US women with or at risk for HIV. *Menopause* 2022;**29**:741–747.
31. Muka T, Oliver-Williams C, Kunutsor S, Laven JSE, Fauser BCJM, Chowdhury R, *et al.* Association of age at onset of menopause and time since onset of menopause with cardiovascular outcomes, intermediate vascular traits, and all-cause mortality: a systematic review and meta-analysis. *JAMA Cardiol* 2016;**1**:767–776.